



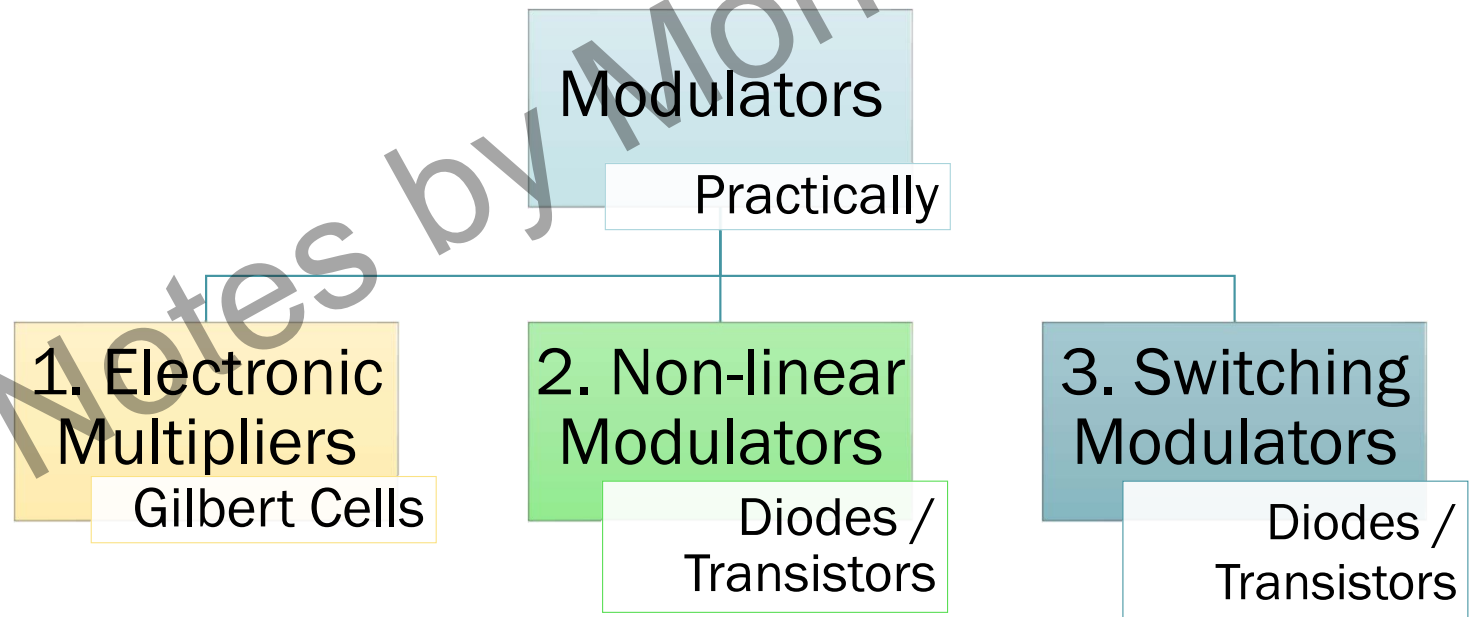
**Ring Modulator**

# Modulator Types

2024/315

# Generation of Amplitude Modulated Signals

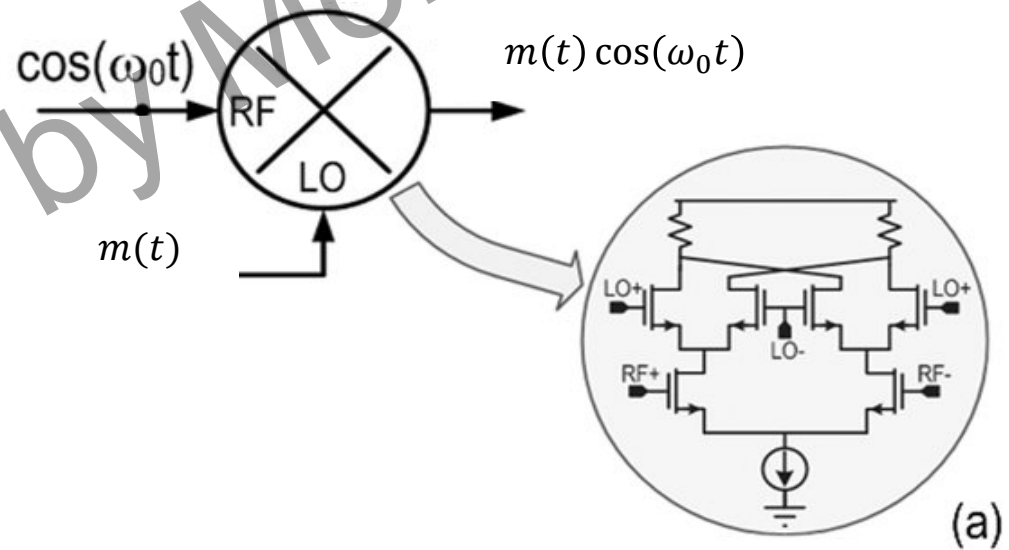
---



# Generation of Amplitude Modulated Signals

## 1. Electronic Multipliers

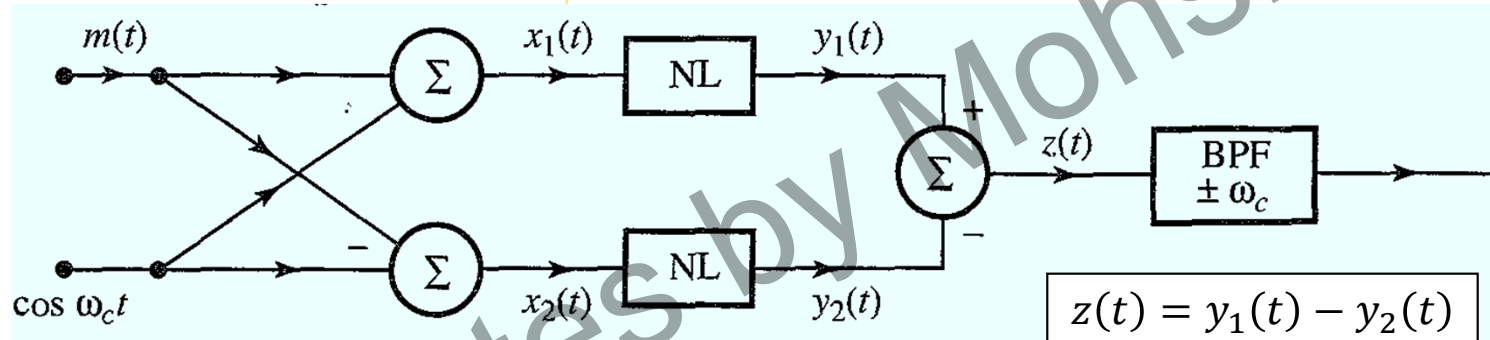
- Modulation is achieved by multiplying  $m(t)$  by  $\cos \omega_c t$  using an analog multiplier whose output is proportional to the product of the two input signals.
  - Difficult to maintain linearity
  - Expensive



# Generation of Amplitude Modulated Signals

## 2. Non-linear Modulators

- Modulation can be achieved by using nonlinear devices, such as a semiconductor diode or a transistor.



$x(t) \rightarrow \text{NL} \rightarrow y(t)$ 
 $y(t) = ax(t) + bx^2(t)$

$$x_1(t) = \cos \omega_c t + m(t)$$

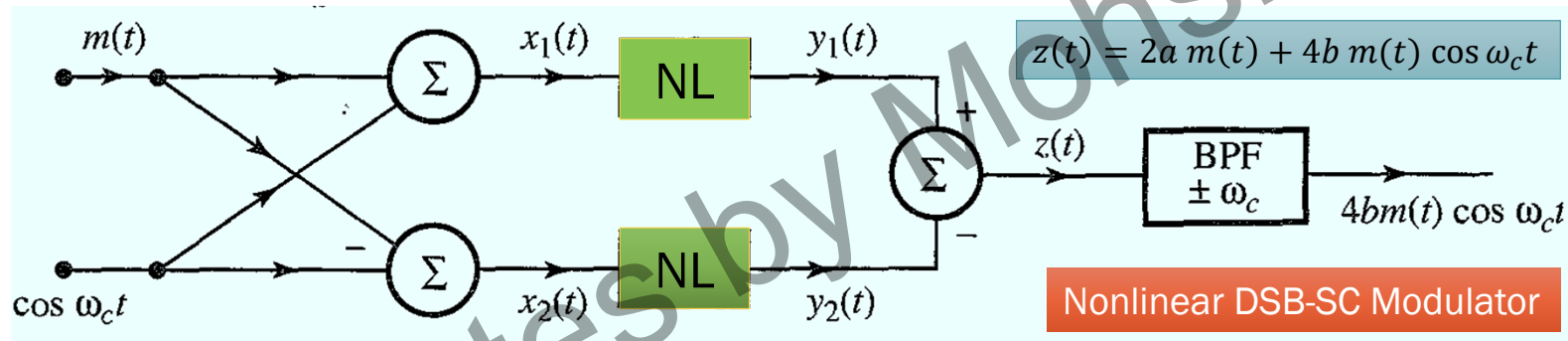
$$x_2(t) = \cos \omega_c t - m(t)$$

$$\begin{aligned}
 y_1(t) &= a[\cos \omega_c t + m(t)] + b[\cos \omega_c t + m(t)]^2 \\
 &= a[\cos \omega_c t + m(t)] + b[\cos^2 \omega_c t + m^2(t) + 2m(t) \cos \omega_c t] \\
 y_2(t) &= a[\cos \omega_c t - m(t)] + b[\cos \omega_c t - m(t)]^2 \\
 &= a[\cos \omega_c t - m(t)] + b[\cos^2 \omega_c t + m^2(t) - 2m(t) \cos \omega_c t]
 \end{aligned}$$

$$z(t) = 2a m(t) + 4b m(t) \cos \omega_c t \rightarrow \text{BPF } \pm \omega_c$$

# Generation of Amplitude Modulated Signals

## 2. Non-linear Modulators



Balanced  
Modulator

Single Balanced

A circuit balanced with respect to only one input.

Nonlinear  
Modulator

Double  
Balanced

A circuit balanced with respect to both inputs.

Ring Modulator

# Generation of Amplitude Modulated Signals

## 3. Switching Modulators

- The multiplication operator required for modulation can be replaced by a simpler switching operation, i.e.  $m(t) \cos \omega_c t$  can be obtained not only by multiplying  $m(t)$  by a pure sinusoid but by **any periodic signal**  $\phi(t)$  of the fundamental radian frequency  $\omega_c$ .
- Trigonometric Fourier series of a **periodic signal** is,

$$\phi(t) = \sum_{n=0}^{\infty} C_n \cos(n\omega_c t + \theta_n)$$

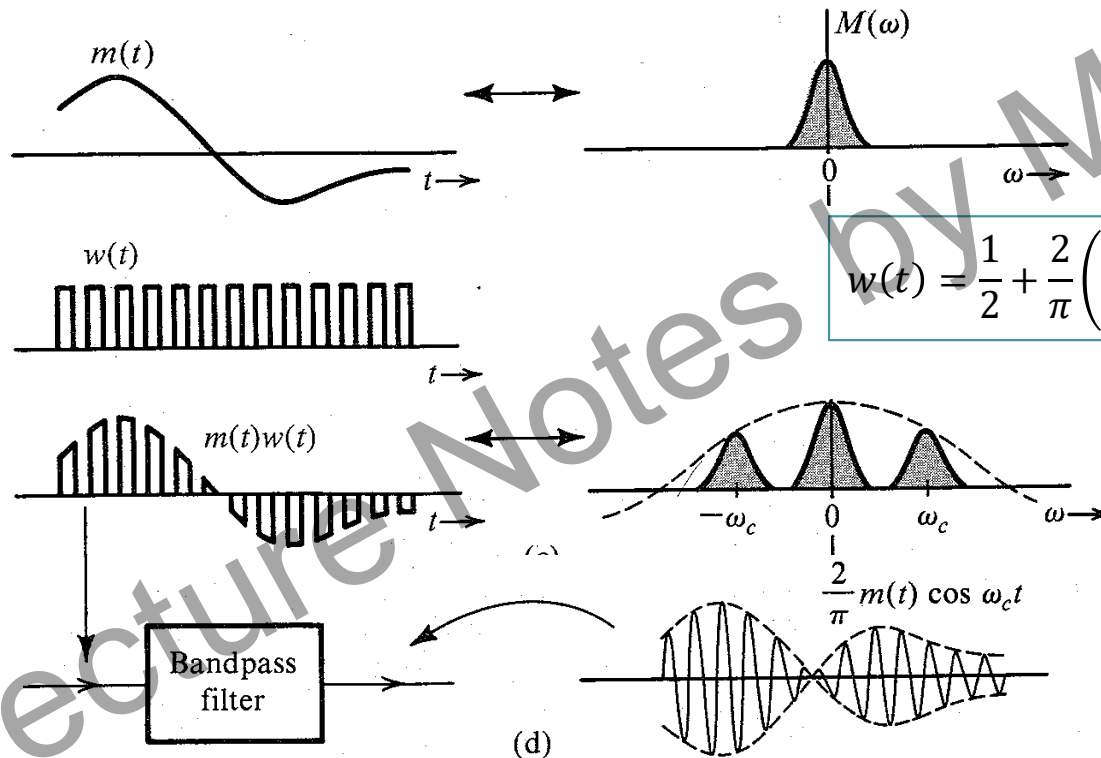
Hence,

$$m(t)\phi(t) = \sum_{n=0}^{\infty} C_n m(t) \cos(n\omega_c t + \theta_n)$$

- ❖ This shows that the spectrum of the product  $m(t)\phi(t)$  is the spectrum  $M(\omega)$  shifted to  $\pm \omega_c, \pm 2\omega_c, \dots, \pm n\omega_c, \dots$ .
- ❖ If this signal is passed through a BPF of bandwidth  $2B$  Hz and tuned to  $\omega_c$ , then we get the desired modulated signal  $C_1 m(t) \cos(\omega_c t + \theta_1)$ .

# Generation of Amplitude Modulated Signals

## 3. Switching Modulators



→ The square pulse train  $w(t)$  is a **periodic signal** whose Fourier series is,

$$w(t) = \frac{1}{2} + \frac{2}{\pi} \left( \cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right)$$

The signal  $m(t)w(t)$  is given by,

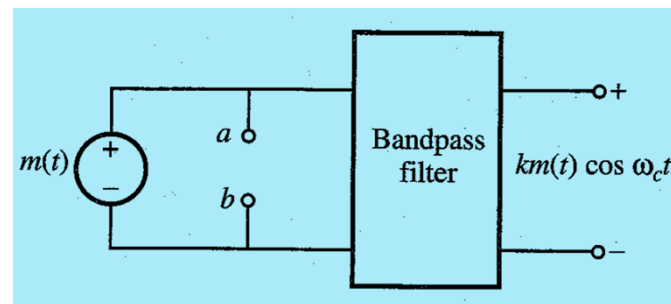
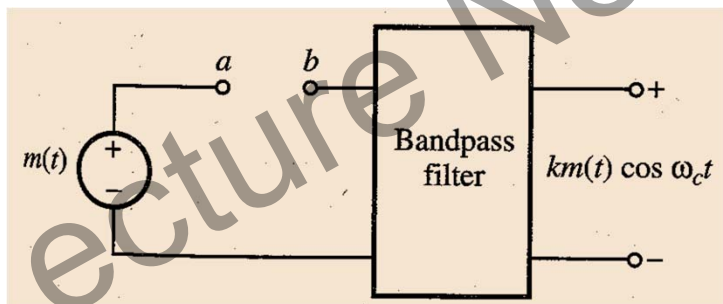
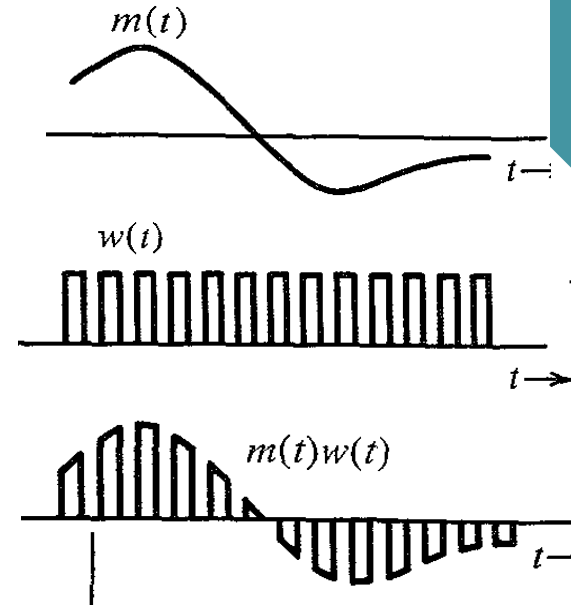
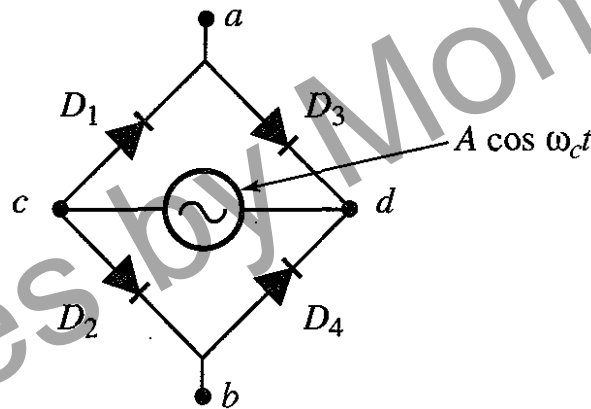
$$\begin{aligned} & \frac{1}{2} m(t) \\ & + \frac{2}{\pi} \left( m(t) \cos \omega_c t - \frac{1}{3} m(t) \cos 3\omega_c t \right. \\ & \left. + \frac{1}{5} m(t) \cos 5\omega_c t - \dots \right) \end{aligned}$$

# Generation of Amplitude Modulated Signals

## 3. Switching Modulators

- Multiplication of signal by a square pulse train is in reality a switching operation. It involves switching the signal  $m(t)$  on and off periodically and can be accomplished by simple switching elements (like diode) controlled by  $w(t)$ .
- To obtain  $m(t)w(t)$ , we may place the switch in series or parallel.

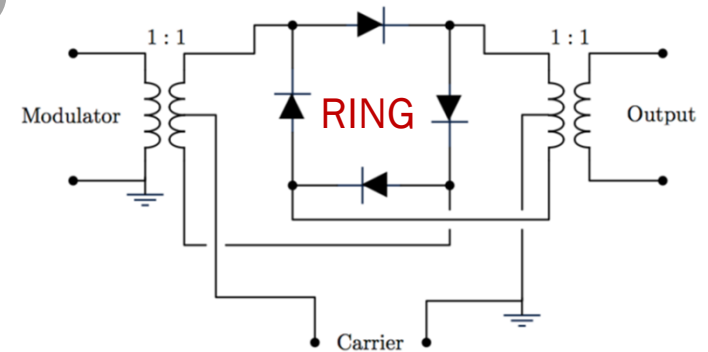
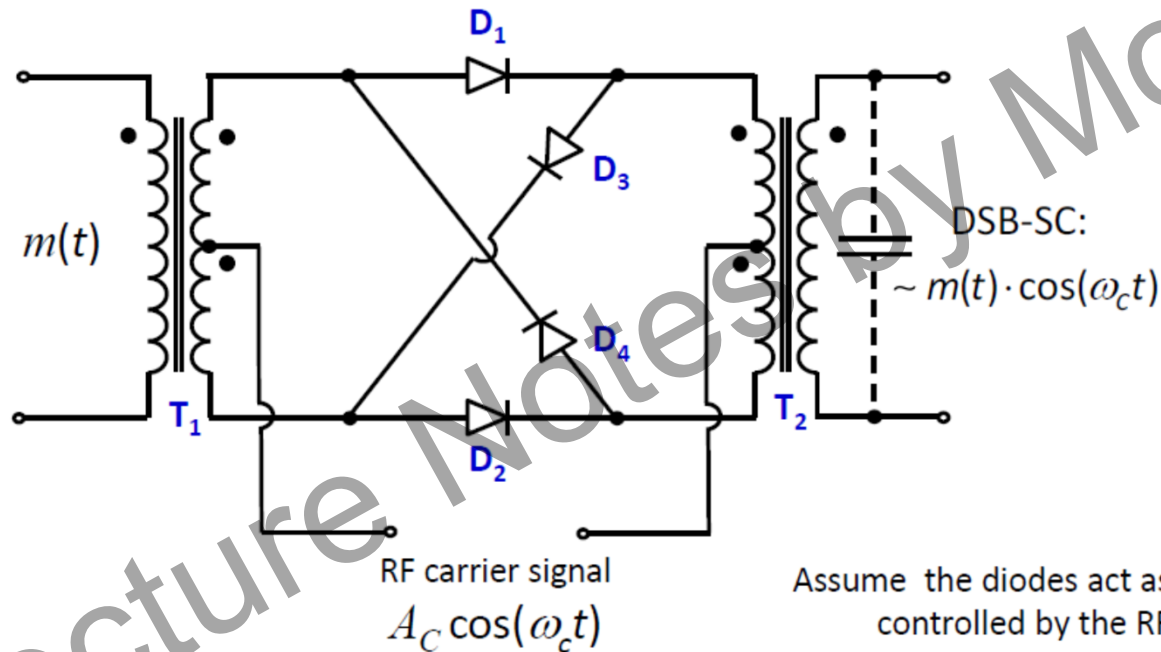
### Diode-bridge Modulator



# Generation of Amplitude Modulated Signals

## 3. Switching Modulators

### Ring Modulator



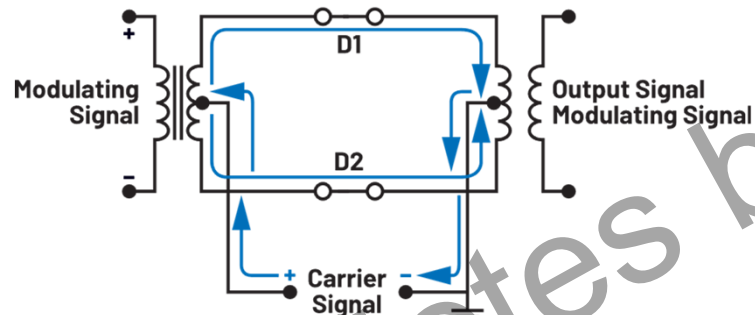
Assume the diodes act as perfect switches (either "on" or "off") and are controlled by the RF carrier signal (requires large amplitude).

# Generation of Amplitude Modulated Signals

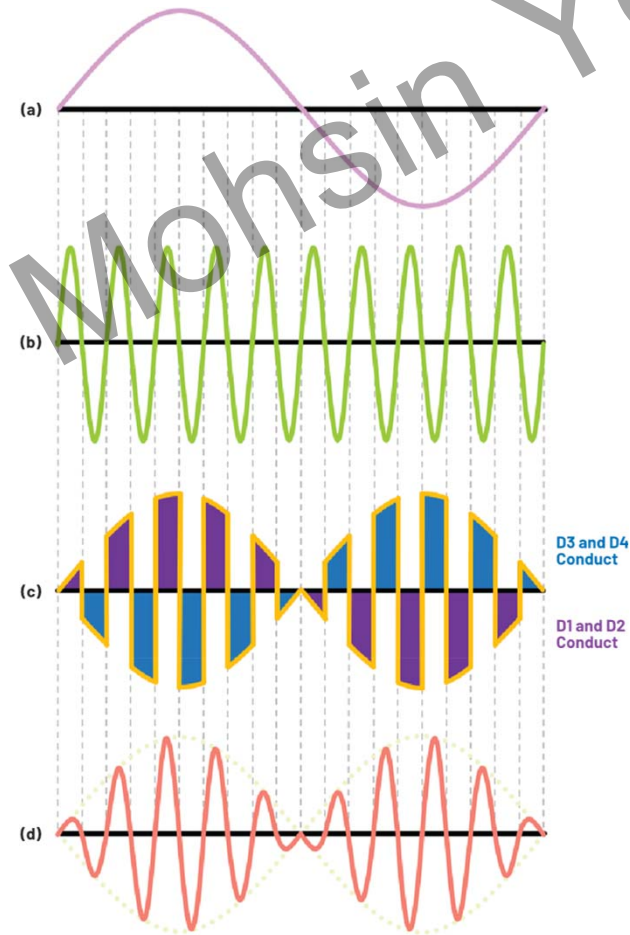
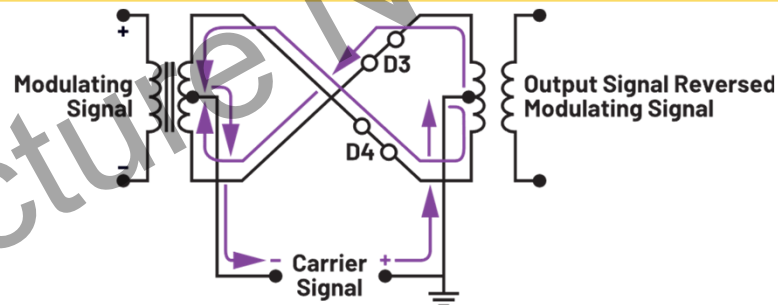
## 3. Switching Modulators

### Ring Modulator

Positive half-cycle of the carrier signal passes  $m(t)$  to output.



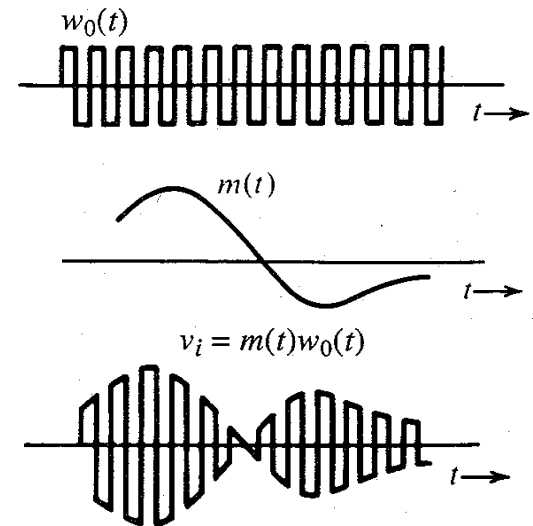
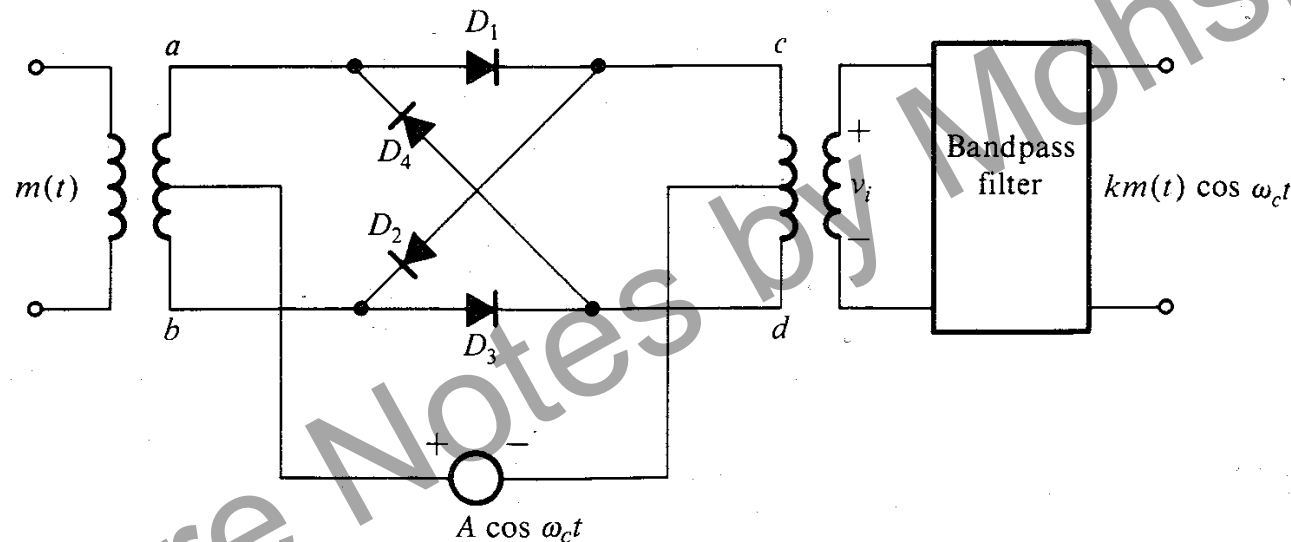
Negative half-cycle of the carrier signal inverts  $m(t)$  at the output.



# Generation of Amplitude Modulated Signals

## 3. Switching Modulators

### Ring Modulator



➤ In effect,  $m(t)$  is multiplied by a square pulse train  $w_0(t)$ , shown in the Figure. The Fourier Series is,

$$w_0(t) = \frac{4}{\pi} \left( \cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right)$$

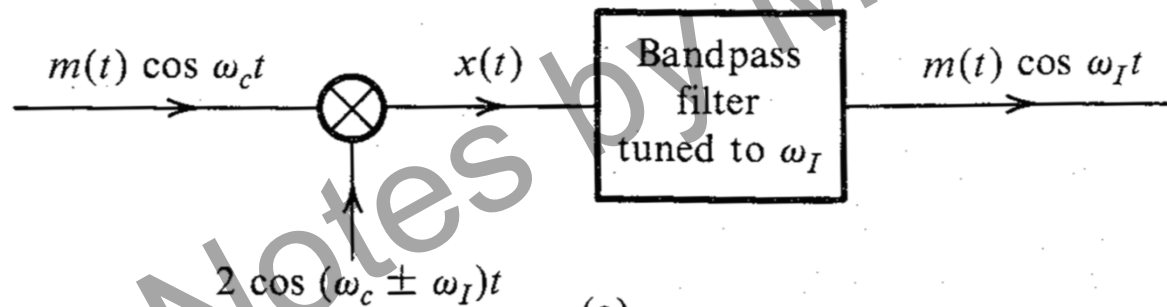
and  $v_i(t)$  is, 
$$v_i(t) = m(t)w_0(t) = \frac{4}{\pi} \left( m(t) \cos \omega_c t - \frac{1}{3} m(t) \cos 3\omega_c t + \dots \right)$$

## Example 4.2 Frequency Mixer / Converter

### Frequency Mixer or Converter

We shall analyze a frequency mixer, or frequency converter, used to change the carrier frequency of a modulated signal  $m(t) \cos \omega_c t$  from  $\omega_c$  to some other frequency  $\omega_I$ .

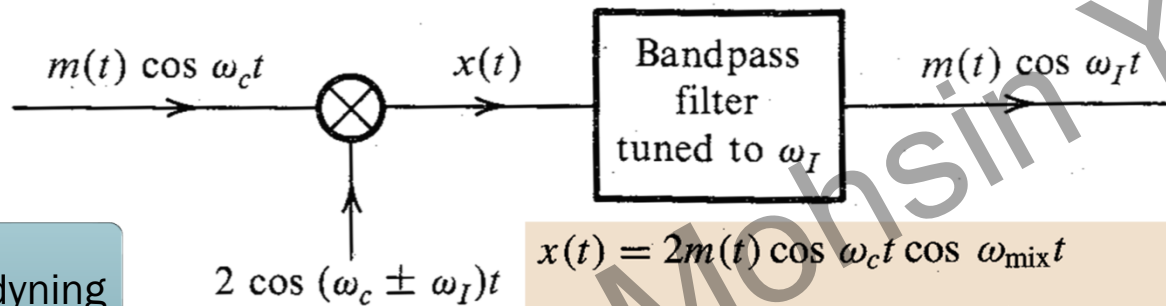
This can be done by multiplying  $m(t) \cos \omega_c t$  by  $2 \cos \omega_{\text{mix}} t$ , where  $\omega_{\text{mix}} = \omega_c + \omega_I$  or  $\omega_c - \omega_I$ , and then bandpass-filtering the product, as shown in Fig. 4.7a.



The product  $x(t)$  is

$$\begin{aligned} x(t) &= 2m(t) \cos \omega_c t \cos \omega_{\text{mix}} t \\ &= m(t) [\cos (\omega_c - \omega_{\text{mix}}) t + \cos (\omega_c + \omega_{\text{mix}}) t] \end{aligned}$$

## Example 4.2 Frequency Mixer / Converter



Heterodyning

Down-Conversion

Up-Conversion

$$x(t) = 2m(t) \cos \omega_c t \cos \omega_{\text{mix}} t$$

$$= m(t) [\cos (\omega_c - \omega_{\text{mix}}) t + \cos (\omega_c + \omega_{\text{mix}}) t]$$

If we select  $\omega_{\text{mix}} = \omega_c - \omega_I$ ,

$$x(t) = m(t) [\cos \omega_I t + \cos (2\omega_c - \omega_I) t]$$

If we select  $\omega_{\text{mix}} = \omega_c + \omega_I$ ,

$$x(t) = m(t) [\cos \omega_I t + \cos (2\omega_c + \omega_I) t]$$

